

Lab 1 Preprocessing

2026-01-15

Week 1 Data Preprocessing

In this week's exercise, we will check our data and prepare it for visualisation.

Goals

- Calculate total distance per month.
- Save into a new csv.

1. Importing data

We can import Martins csv, or use the following code to download the same dataset.

```
file_id <- "1ymbNqfv9s6YGZzN93HFKAhjg0Z5xZXV1"
url <- paste0("https://drive.google.com/uc?id=", file_id)

df <- read.csv(url)
```

2. Saving to CSV

If we downloaded the data we can re-save to csv with;

```
write.csv(df, "runningdata.csv", row.names = FALSE)
```

If you are unsure where the file is saved to you can check the working directory with;

```
getwd()
```

You can open and investigate the csv in excel, text editor or spreadsheet program of your choice.

3. Explore the dataframe in the viewer

Double click df in the environment pane usually on the right to open the data in the viewer

!bardata

4. Get the column headers programatically.

```
## [1] "achievement_count"      "athlete"
## [3] "athlete_count"          "average_cadence"
## [5] "average_heartrate"      "average_speed"
## [7] "average_temp"           "average_watts"
## [9] "comment_count"          "commute"
## [11] "device_watts"           "display_hide_heartrate_option"
## [13] "distance"               "elapsed_time"
## [15] "elev_high"              "elev_low"
## [17] "end_latlng"             "external_id"
## [19] "flagged"                "from_accepted_tag"
## [21] "gear_id"                "has_heartrate"
## [23] "has_kudoed"             "heartrate_opt_out"
## [25] "id"                     "kilojoules"
## [27] "kudos_count"            "location_city"
## [29] "location_country"       "location_state"
## [31] "manual"                 "map"
## [33] "max_heartrate"          "max_speed"
## [35] "max_watts"              "moving_time"
## [37] "name"                   "photo_count"
## [39] "pr_count"               "private"
## [41] "resource_state"          "sport_type"
## [43] "start_date"             "start_date_local"
## [45] "start_latlng"           "suffer_score"
## [47] "timezone"               "total_elevation_gain"
## [49] "total_photo_count"      "trainer"
## [51] "type"                   "upload_id"
## [53] "upload_id_str"          "utc_offset"
## [55] "visibility"             "weighted_average_watts"
## [57] "workout_type"
```

5. Create a new column containing only the year and month from the date data.

You have choices in how to do this, I recommend you avoid packages at this point and use `as.POSIXct`. <https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/as.POSIXct>

You can follow the previous step with `format`. <https://www.geeksforgeeks.org/r-language/how-to-use-date-formats-in-r/>

```
## [1] "2025-07" "2025-07" "2025-07" "2025-07" "2025-07" "2025-07"
```

6. Aggregate data

Using `aggregate(response ~ group, data = df, FUN = ...)`, create a new dataframe that groups runs by YearMonth and sums Distance to calculate the total distance run each month.

```
##   YearMonth distance
## 1  2011-01  11251.0
## 2  2011-02   8810.9
## 3  2011-03   5032.1
## 4  2011-04   6367.8
## 5  2011-07  32238.8
## 6  2011-08  23952.2
```

7. Save data to csv

```
write.csv(monthly, "monthly_stats.csv", row.names = FALSE)
```